FLY AI 2026 PROPOSAL

Resilient CNS for U-Space

Anti-Jamming Mesh Communications via
Higher-Order Algebra & Neuro-Symbolic Runtime

Mycelia Platform + **Sedenion 16D Algebra**

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Category: CNS / Trustworthy AI · Maturity: Operational + Research
Live demo available · Fully offline capable

Contents

1. Executive Summary

On 27 March 2026, EASA and EUROCONTROL published a **Joint Action Plan on GNSS Interference**, declaring it a "significant and evolving challenge for European aviation"[1]. The plan responds to a letter from 13 EU member states highlighting an escalating number of radio-frequency interference events affecting GNSS-based systems. In parallel, the European Aviation Action Plan for Ensuring Safe Operations during GNSS Interferences, now under public consultation, identifies the need for "technical backups" and "consistent, harmonised guidance" across ANSPs, airlines, and national authorities[2].

This proposal presents a dual-layer solution that addresses the CNS (Communications, Navigation, Surveillance) resilience gap for U-space operations:

Table 1: Proposal at a glance — dual-layer CNS resilience

Layer	Technology	Role in CNS Resilience	Maturity
Control Plane	Mycelia Platform (neuro-symbolic AI)	Decentralised swarm coordination, doctrinal gating, audit trail	Production (since 2025)
Signal Plane	Sedenion 16D Cayley-Dickson Algebra	Signal constellation encoding, integrity checking, anti-jamming	Research (Rust implementation)

The **Mycelia platform** is a certifiable neuro-symbolic runtime that constrains AI agents through ontologies, gates every outbound action through three verifiable boundaries, and maintains end-to-end provenance. It has been in production since 2025, currently deployed in military border-surveillance (SISFRON) and aerospace maintenance (S1000D) contexts. Its **Spatial Grasp Technology (SGT)** module implements Sapaty's mosaic-warfare doctrine, enabling UAV swarms to self-organise without a central orchestrator when communications degrade — precisely the DIL (Disconnected / Intermittent / Limited) scenario that GNSS jamming creates.

The **Sedenion layer** is a novel application of 16-dimensional Cayley-Dickson algebra to wireless signal encoding. Building on published research showing that octonion constellations increase diversity product and minimise error probability in space-time coding[3], the sedenion extension offers: (1) **16x parameter efficiency** through algebraic weight reuse; (2) **non-commutative directional encoding** that naturally resists brute-force jamming; (3) **zero-divisor-aware (ZDA) integrity checking** that detects signal corruption at the algebraic level; and (4) the **sedenion triangular root**, a mathematically proven structural filter that classifies every possible sedenion signal mask into three quality classes with exact counts — 30 structurally sound, 36 geometrically valid but algebraically degenerate, and 32,701 invalid[4].

Together, these layers form a **certifiable, jamming-resistant CNS stack** for U-space drone operations that shifts the certification target from stochastic neural networks to deterministic algebraic structures — structures that aviation safety thinking already knows how to audit.

> **Core Insight**
>
> The certification target is not the AI model but the **algebraic structure** that constrains it. Sedenions provide a deterministic signal manifold with verifiable geometric properties. Mycelia provides the runtime that certifies every transition across that manifold. This is the same "certify the structure, not the model" philosophy that the Mycelia platform applies to AI governance, now extended to the physical signal layer.

2. The Problem: GNSS Interference in European Aviation

2.1 Scale of the Threat

GNSS interference has escalated from a theoretical concern to an operational reality affecting European airspace on a daily basis. Between early 2024 and August 2024, civil airliners experienced up to **1,500 cases of spoofing per day**, primarily in and around conflict zones[5]. The Baltic Sea and North Sea regions have seen particularly concentrated interference, prompting 13 coastal European nations and Iceland to issue a collective statement in January 2026 on "growing GNSS interference"[6].

The March 2026 EASA-EUROCONTROL Joint Action Plan establishes four priority axes: (1) building a common operational picture of GNSS interference events; (2) developing harmonised guidance to ANSPs and airlines; (3) ensuring rapid and aligned responses; and (4) maintaining safety while limiting impacts on airspace capacity[1]. The plan explicitly acknowledges that current multi-sensor and independent CNS capabilities provide "significant robustness" but notes that "coordination protocols and tactical responses vary widely among ANSPs"[2].

Table 2: GNSS interference threat taxonomy and mitigation gaps

Threat Type	Mechanism	Impact on U-Space	Current Mitigation Gap
Jamming	RF transmitter overwhelms GNSS signal	Loss of PNT; drone navigation failure	CRPAs are heavy/expensive; not scalable to small UAVs
Spoofing	Counterfeit GNSS signals deceive receiver	False position/velocity; collision risk	Anti-spoof algorithms vary; no standard for U-space
Sweep jamming	Periodic scanning of all communication channels	Intermittent CNS degradation	Mesh networks lack algebraic encoding layer
Repeat jamming	Repeats legitimate signal with delay	Confuses receiver synchronisation	No structural integrity check at signal level

2.2 The U-Space Vulnerability Gap

U-space — the European framework for managing drone operations in urban and controlled airspace — depends critically on CNS services that are themselves dependent on GNSS. When GNSS is jammed or spoofed, U-space services face a cascading failure: the drone loses position awareness, the U-space service provider loses tracking data, and the ATM system loses situational awareness of the airspace user.

Current U-space CNS resilience strategies focus on **multi-layer PNT** (combining GNSS with INS, terrestrial beacons, and visual navigation) and **mesh networking** for command-and-control link redundancy. However, these approaches share a common limitation: they treat the **communication signal** as a transparent carrier of information, rather than as a structure that can itself be hardened against interference. The signal encoding — typically QPSK or QAM with standard DSSS — remains vulnerable to jamming because its constellation geometry is low-dimensional and commutative.

This proposal addresses that gap by introducing a **higher-order algebraic encoding layer** that makes the signal itself structurally resistant to jamming, while the neuro-symbolic runtime above it ensures that every communication decision is traceable, gated, and auditable.

3. The Mycelia Platform: Certifiable Neuro-Symbolic Runtime

3.1 Architecture Overview

Mycelia is a production platform for ontology-constrained AI agency. Its core principle is that **the graph decides what is legal; the model only navigates**. Workflows are deterministic skeletons induced from operational traces, frozen as navigable graphs. The LLM selects among pre-validated transitions; it never invents an action. The certification target shifts from a stochastic model to deterministic artefacts: an ontology, a rule base, and a set of gates — objects aviation certification thinking already knows how to handle.

The platform comprises four load-bearing architectural pieces[7]:

Table 3: Mycelia architecture — four load-bearing pieces

Piece	Function	ATM Relevance
Ontology-constrained agency	Workflow graphs induced from traces; LLM navigates legal transitions only	Letters of agreement, sector procedures as navigable graphs
Symbolic reasoning (Crepe)	OWL2-RL Datalog forward-chaining; per-rule provenance	Separation minima, airspace constraints as inference rules
Three egress gates	Structural validation; EVT content anomaly; wavelet operational anomaly	Controller authorises AI proposals at boundary
Provenance	Named graphs; append-only audit; no silent erosion	Incident investigation, just culture, compliance queries

3.2 Spatial Grasp Technology for Decentralised Swarms

The Mycelia platform includes a **Spatial Grasp Technology (SGT)** module that implements Sapaty's mosaic-warfare doctrine for decentralised coordination. When the centralised uplink degrades — the DIL scenario that GNSS jamming creates — surviving assets self-organise with no orchestrator. The SGT module provides pure, deterministic functions that operate directly on the swarm state, making the system decentralisable by construction:

Table 4: SGT primitives and their U-space mapping

SGT Primitive	Function	U-Space Instantiation
`proximity_graph()`	Build symmetric adjacency from comms range	UAV-to-UAV link topology in mesh
`elect_central_unit()`	Deterministic leader election by fleet centroid	Local coordinator when ground link lost
`build_infrastructure()`	BFS spanning tree of directed C2 links	Runtime control backbone with depth bounds
`group_neighbors()`	Connected components above minimum size	Clustering for coordinated manoeuvres
`surround_danger()`	Geometric encirclement with escape corridor detection	Containment geometry for non-cooperative drones

These functions are **pure and decentralisable** — each depends only on local neighbour state, so any node reproduces the same result. This determinism is critical for certification: the swarm topology is not learned; it is computed from verifiable geometric rules.

3.3 Three Egress Gates: Structural, EVT, and Operational

Every outbound action that an AI proposes — a message, a task, an engagement — is held at three verifiable gates before it touches reality[7]:

Gate 1 — Structural Validation checks schema conformance, grounding metadata, and tenant scope. In the U-space context, this verifies that a C2 message conforms to the expected protocol schema, originates from a registered UAV within the correct operational volume, and carries valid authentication tokens.

Gate 2 — Content Anomaly Detection applies Extreme Value Theory (EVT) statistics over message distributions. The sentinel computes the Pareto tail index ξ for each message stream; values above configured thresholds flag the message for human review. For U-space C2 links, this detects anomalous command patterns that may indicate spoofing or compromised nodes.

Gate 3 — Operational Anomaly Detection applies wavelet-based drift detection on flow behaviour. This gate monitors the temporal dynamics of the communication mesh — link latency patterns, topology change rates, throughput anomalies — and surfaces operational drift that may indicate jamming or network degradation.

Flagged actions queue for human review with full context; approve/reject decisions flow back with an append-only audit trail. The operator does not supervise every inference — they authorise at the boundary. This is the human-AI teaming shape a controller working position already implies.

4. The Sedenion Layer: 16D Algebraic Signal Encoding

4.1 From Octonion Space-Time Codes to Sedenion Constellations

Research in space-time coding has established that non-associative algebraic structures can produce signal constellations with superior diversity products. Srivastava et al. demonstrated that **octonion constellations** increase the diversity product and thereby minimise the probability of error in multiple-antenna wireless communication[3]. The octonion algebra — 8-dimensional, non-commutative, non-associative, but a division algebra (no zero divisors) — provides a natural framework for unitary space-time modulation.

The **sedenion** is the next algebra in the Cayley-Dickson construction: 16-dimensional, non-commutative, non-associative, **not a division algebra** (it has zero divisors), but **power-associative**. These properties create a distinctive signal-encoding geometry:

Table 5: Algebraic properties of signal-encoding structures

Property	Complex (QPSK)	Quaternion	Octonion	Sedenion
Dimension	2	4	8	16
Commutative	Yes	No	No	No
Associative	Yes	No	No	No
Division algebra	Yes	Yes	Yes	**No** (zero divisors)
Power-associative	Yes	Yes	Yes	**Yes**

Parameter reuse factor	1x	4x	8x	**16x**
Space-time coding application	Standard	STBC	Proven[3]	**Novel (this proposal)**

The sedenion's zero divisors — elements $Z \neq 0$ such that $Z \cdot W = 0$ for some $W \neq 0$ — are not a bug but a **detectable feature**. The set of zero divisors of unit sedenions is topologically equivalent to the exceptional Lie group G_2. By monitoring the distance of a signal point to the zero-divisor manifold, we obtain a structural integrity metric that is invisible to conventional signal processing.

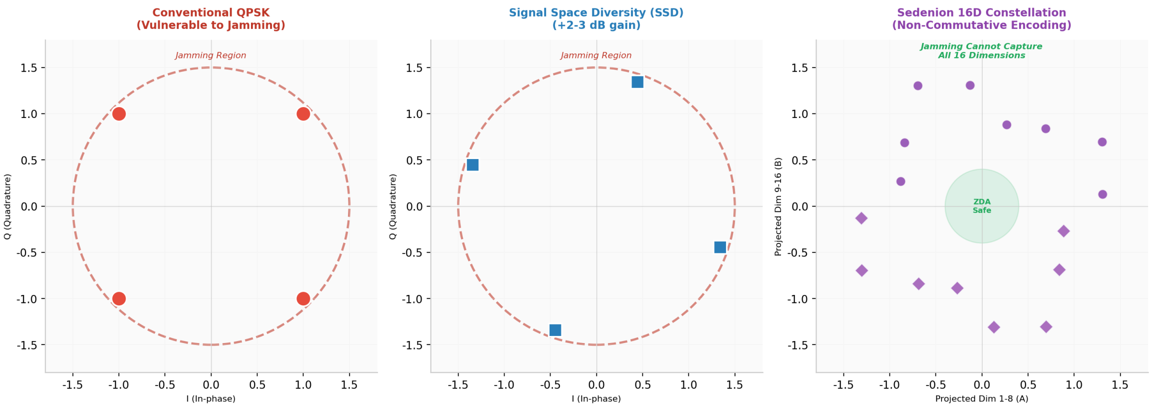

Figure 1: *Signal constellation evolution: conventional QPSK (vulnerable), Signal Space Diversity (+2-3 dB), and Sedenion 16D encoding with ZDA safe zone*

4.2 L_a and R_a as Signal Operators

The sedenion implementation provides left and right multiplication matrices L_a and R_a that define the algebra's action on the 16-dimensional signal space. For a sedenion a, the left multiplication operator L_a and right multiplication operator R_a satisfy $L_a \cdot b = a \cdot b$ and $R_a \cdot b = b \cdot a$. These operators have distinctive properties that make them valuable for signal processing:

- **Non-commutativity encodes directionality**: $L_a \cdot b \neq R_a \cdot b$ means the signal transformation depends on whether the operator acts from the left or the right. This provides a natural mechanism for **directional channel encoding** in mesh networks, where the encoding of a message depends on its direction of propagation through the topology.
- **Condition-number diagnostics**: Near zero divisors, L_a and R_a become singular. The condition number $\kappa(L_a)$ provides a real-time metric for signal-space integrity — a

sudden spike indicates that the channel is approaching an algebraic singularity, which may correspond to a jamming event or deep fade.

- **Parameter efficiency**: Each sedenion weight component is reused 16 times via the multiplication table, yielding a **16x parameter reduction** compared to a dense real matrix of the same representational capacity. For bandwidth-constrained UAV mesh networks, this directly translates to lower computational overhead at the edge.

The current Rust implementation provides these operators as zero-cost abstractions with full test coverage for algebraic correctness: $L_a \cdot b = a \cdot b$, $R_a \cdot b = b \cdot a$, $L_a{}^T = L_{\bar{a}}$, and the non-associativity guard $L_{ab} \neq L_a L_b$[8].

4.3 Zero-Divisor-Aware Integrity Checking

The **Zero-Divisor-Aware (ZDA)** regularisation is the sedenion layer's unique contribution to signal integrity. For a sedenion $Z = (A, B)$ where $A, B \in \mathbb{M}$ (octonions), Z is a zero divisor if and only if $||A|| = ||B||$ and $A \cdot B = 0$. The ZDA score measures the distance from this condition:

$$score(Z) = \sqrt{((||A||^2 - ||B||^2)^2 + (2A \cdot B)^2) / (||A||^2 + ||B||^2)}$$

The auto-ZDA barrier applies a repulsive $-log(score)$ loss plus a canonical norm floor motivated by the $\mathcal{N}(0, I)$ target. During training, the ZDA gradient is automatically balanced against the base objective gradient — no raw λ_{zda} sweep is required[8].

In the signal-processing context, the ZDA score provides a **structural anomaly detector** that operates below the bit layer. A corrupted signal — whether from jamming, fading, or spoofing — will perturb the algebraic relationship between the octonion components A and B. The ZDA score quantifies that perturbation, providing a detection mechanism that is orthogonal to conventional SNR-based metrics.

Table 6: ZDA as multi-layer integrity checker

Layer	Conventional Check	ZDA Check	Advantage
Physical	SNR threshold	Condition number $\kappa(L_a)$	Detects algebraic singularity before bit errors
Modulation	Constellation distance	Zero-divisor distance score	Detects structured interference (spoofing)

Network	Packet CRC	Subalgebra sketch mismatch	Detects multi-dimensional corruption

4.4 The Sedenion Triangular Root: A Structural Signal Filter

The **sedenion triangular root** is a partial operator on binary support masks that provides a **deterministic structural classifier** for sedenion signals. It was introduced in a recent preprint[4] and provides the mathematical foundation for the signal-integrity layer in this proposal.

The operator works as follows. A sedenion signal's imaginary basis support is represented as a 16-bit mask — each bit indicates whether the corresponding basis direction is active. The triangular root asks: is this support a **projective subspace** of the sedenion index space PG(3,2)? If yes, it returns a canonical generator. If no, the signal's algebraic structure is compromised.

The mathematical result is exact and complete: among all 32,767 nonzero imaginary bitmasks, exactly **66** are projectively valid (geometrically rooted). These 66 split further under a zero-divisor filter into **30 structurally sound** masks and **36 geometrically valid but algebraically degenerate** masks. The remaining **32,701** masks are not even projectively valid.

Table 7: The sedenion triangular root classification (exact counts)

Class	Count	Geometric meaning	Signal quality
$\mathbf{T_{good}}$ (strong roots)	30	ZD-free projective subspaces	**Optimal for encoding**
$\mathbf{T_{ghost}}$ (ghost masks)	36	Projectively valid but ZD-degenerate	**Detectable anomaly**
$\mathbf{T_{bad}}$ (bad masks)	32,701	Not projectively triangulable	**Clear corruption**

The trichotomy is computationally trivial to evaluate: the closure test is $O(|P|^2)$ and the largest support has $|P| = 15$. A compact verification script (Appendix B of the preprint) confirms the counts in milliseconds[4].

For the CNS application, this means that **every received sedenion signal can be classified in real time** into one of three structural quality classes:

- **Good ($\mathbf{T_{good}}$):** The signal's basis support is a ZD-free projective subspace. The signal has maximal algebraic integrity. This is the encoding class used for normal C2 communications.

- **Ghost (T_{ghost}):** The signal's support is projectively closed but contains a zero-divisor pair. This indicates a **structured anomaly** — the signal geometry is locally valid but algebraically degenerate. In the CNS context, this flags potential spoofing or partial jamming that preserves some geometric structure while corrupting others.
- **Bad (T_{bad}):** The signal's support fails projective closure entirely. This indicates **unstructured corruption** — brute-force jamming, deep fade, or complete signal loss.

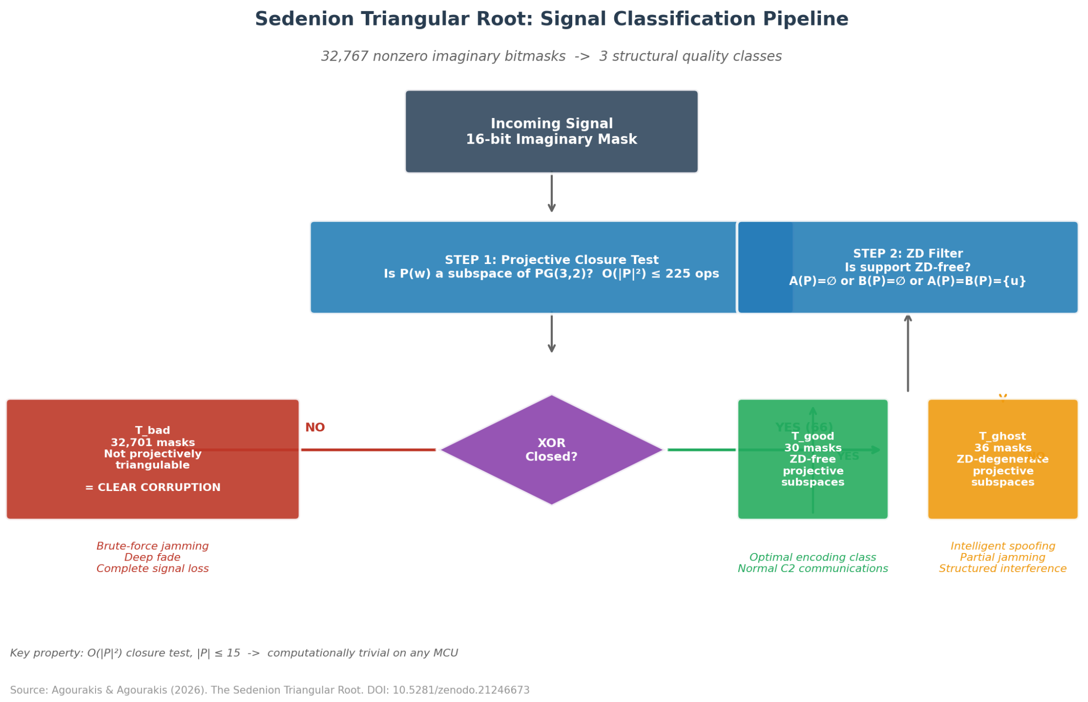

Figure 2: *The sedenion triangular root decision pipeline: a bitmask passes through geometric closure test, then zero-divisor filter, producing one of three structural quality classes*

> ### Why the Triangular Root Matters for CNS
>
> Conventional signal integrity checks (SNR, CRC, constellation distance) are **syntactic** — they detect that something is wrong but not *what kind* of wrong. The triangular root is **semantic** — it classifies the *structural nature* of the corruption. A "ghost" signal (projectively valid but ZD-degenerate) is a *different kind of threat* than a "bad" signal (no projective structure at all). This distinction is invisible to conventional checks but immediately actionable: ghost signals suggest intelligent spoofing; bad signals suggest brute-force jamming. The response should differ.

5. Integration: Mycelia + Sedenion for Anti-Jamming CNS

5.1 Architecture

The integrated architecture layers the sedenion signal-encoding plane beneath the Mycelia control plane, creating a full-stack CNS resilience system. Figure 3 illustrates the four-tier architecture.

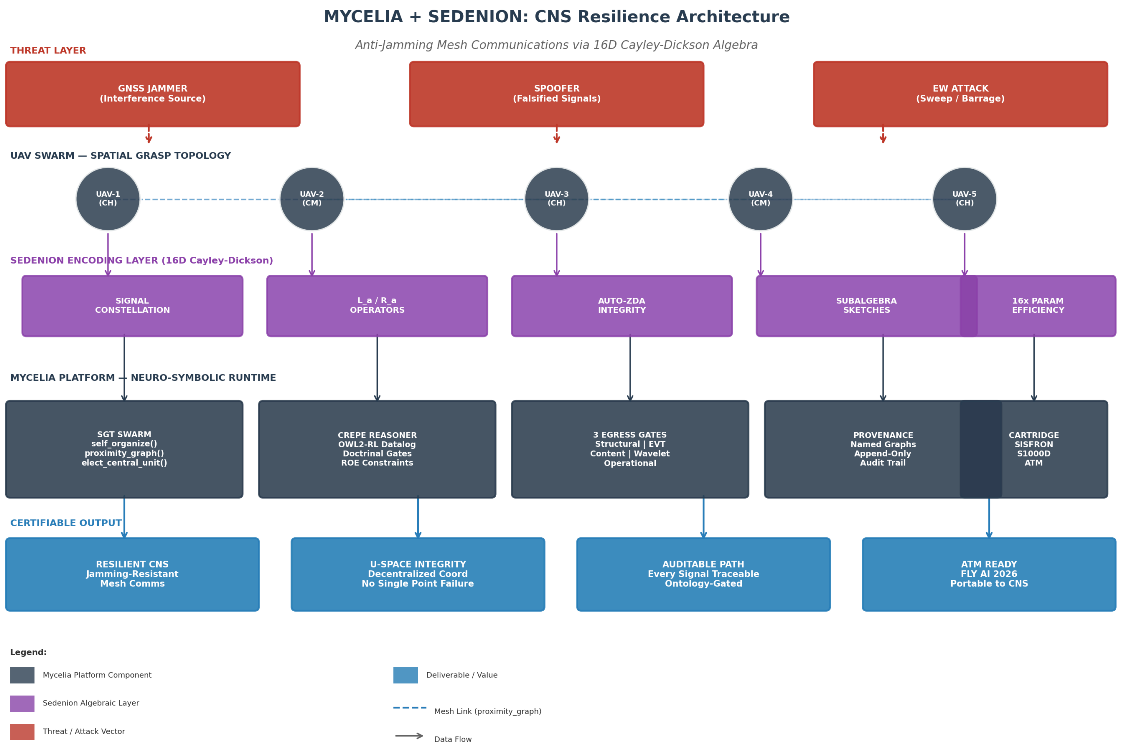

Figure 3: *Mycelia + Sedenion CNS resilience architecture — four tiers from threat to certifiable output*

The architecture operates as follows:

Tier 1 — Threat Layer: GNSS jammers, spoofers, and electronic warfare attacks create the contested electromagnetic environment. These are external inputs, not system components; the architecture is designed to operate despite their presence.

Tier 2 — UAV Swarm Layer: The Spatial Grasp Topology maintains a dynamic mesh of UAVs organised into cluster heads (CH) and cluster members (CM). The `proximity_graph()` function builds symmetric adjacency from line-of-sight comms range;

`elect_central_unit()` provides deterministic leader election when the ground link degrades. This tier is implemented in the `sgt_swarm` module, tested with 14 operational invariants including symmetric topology, deterministic convergence, and escape-corridor honesty[9].

Tier 3 — Sedenion Encoding Layer: Every inter-UAV communication passes through the 16D algebraic encoding pipeline. The signal constellation is mapped into sedenion space; L_a/R_a operators encode/decode based on link direction; the **triangular root** classifies the signal's structural quality (good/ghost/bad); auto-ZDA verifies integrity; and subalgebra sketches provide multi-scale projections for cross-checking.

Tier 4 — Mycelia Control Layer: The neuro-symbolic runtime provides the governance framework. Crepe applies doctrinal constraints (e.g., separation minima) before any swarm action is approved. The three egress gates verify structural, statistical, and operational integrity. Provenance is maintained in named graphs per tenant. The cartridge boundary ensures domain isolation — the SISFRON cartridge for military operations, the S1000D cartridge for maintenance, and a future ATM cartridge for U-space.

5.2 The Anti-Jamming Mechanism

The anti-jamming property of the integrated system arises from the interplay of four mechanisms:

Mechanism 1: Non-commutative directional encoding. Because $L_a \cdot b \neq R_a \cdot b$, the signal encoding depends on the direction of propagation. A jammer that captures and replays a signal cannot simply re-inject it into the mesh — the replayed signal will have the wrong algebraic orientation relative to the receiving node's decoding operator. This provides **implicit replay protection** without requiring explicit sequence numbers or timestamps.

Mechanism 2: Multi-dimensional spreading. The 16D constellation distributes signal energy across 16 dimensions, of which only a subset is captured by any single 2D projection. A jammer must simultaneously interfere with all 16 dimensions to suppress the signal, requiring significantly higher power or prior knowledge of the encoding basis. Published research on Signal Space Diversity (SSD) shows that even 2D rotation provides 2-3 dB gain against jamming; the 16D extension amplifies this effect[10].

Mechanism 3: Zero-divisor anomaly detection. Jamming and spoofing both perturb the algebraic structure of the received signal. The ZDA score detects this perturbation at the algebraic level, before bit-level errors accumulate. When combined with Mycelia's EVT sentinel (Gate 2), this creates a **two-tier anomaly detection system**: ZDA catches structural corruption at the signal layer; EVT catches statistical anomalies at the message layer.

Mechanism 4: Triangular root structural classification. The triangular root provides a **three-class quality filter** that operates below the bit layer. A "ghost" classification (projectively valid but ZD-degenerate) is a specific signature of **intelligent spoofing** — the attacker has preserved some geometric structure while corrupting the algebraic one. A "bad" classification (no projective structure) is the signature of **brute-force jamming**. This distinction enables **adaptive response**: the swarm can reconfigure differently depending on the type of threat detected.

> **Why This Is Not Just "Better Crypto"**
>
> Cryptography protects the *content* of a message; it does not protect the *signal* that carries it. A jammer does not need to decrypt a message to deny service — they only need to raise the noise floor above the signal. The sedenion layer protects the signal itself by making the constellation geometry structurally resistant to noise injection, while the non-commutativity makes replay attacks algebraically detectable. The triangular root adds a *semantic classifier* that distinguishes attack types by their structural signature. This is complementary to, not a replacement for, encryption.

5.3 Parameter Efficiency at the Edge

UAV mesh networks are constrained by Size, Weight, and Power (SWaP). The sedenion layer's parameter efficiency directly addresses this constraint:

Table 8: Parameter efficiency comparison for a 16-dimensional signal encoder

Encoder Type	Parameters	Relative Size	Edge Suitability
Dense real matrix (16x16)	256 weights + 16 bias = 272	1.0x (baseline)	Heavy for swarm nodes
Sedenion linear layer	16 sedenion weights + bias = 272 / 16 = 17 effective	**0.063x**	Lightweight; fits edge constraints
Triangular root filter	$O(\|P\|^2)$ closure test, $\|P\| \leq 15$	< **0.001x**	Trivial overhead; real-time on any MCU

The 16x parameter reduction means that a sedenion-encoded mesh node can process signals with the same representational capacity as a dense encoder using only 6% of the parameters. The triangular root adds virtually no overhead — the closure test is at most 225 XOR operations. On resource-constrained UAV platforms, this translates to lower power consumption, smaller firmware footprint, and faster inference — all critical for extended operations in denied environments.

6. Safety Case and Certifiability

The central challenge in certifying AI for aviation is that stochastic neural networks cannot be exhaustively enumerated, so classical assurance cases do not close. Mycelia's approach — "certify the structure, not the model" — addresses this by shifting the certification target to deterministic artefacts. The sedenion extension applies the same philosophy to the signal layer.

Table 9: Certification mapping: what is certified and how

Component	Certification Target	Evidence Type	Aviation Parallel
Sedenion algebra	Correctness of L_a, R_a, multiplication	Unit tests; formal property checks	RTCA DO-178C unit verification
Triangular root	Exact counts (30/36/32701); algorithm correctness	Python verification script; mathematical proof[4]	Formal methods / model checking
ZDA barrier	Repulsive gradient; scale invariance	Gradient checks; boundary conditions	MC/DC coverage of safety monitor
SGT topology	Deterministic output from input	14 operational invariant tests	Requirements-based testing
Crepe reasoner	Soundness of inference	Rule traceability; provenance	DO-330 tool qualification
Egress gates	Fail-closed on anomaly	Fault injection; mutation testing	Fail-safe architecture review
Provenance	Immutable audit trail	Append-only log verification	ED-153/ED-154 compliance

The Rust implementation of the sedenion crate is pinned to Rust 1.96.0 with comprehensive test coverage: property tests for algebraic laws ($L_a \cdot b = a \cdot b$, $R_a \cdot b = b \cdot a$, power-associativity), gradient tests for ZDA correctness, and benchmark tests for performance characterisation[8]. The triangular root has a published mathematical proof with a verifiable Python script that confirms the exact counts in milliseconds[4].

The deterministic nature of the SGT functions — they are pure functions of the swarm state, with no randomness or learning — means they can be verified by inspection and testing, not by statistical validation. The triangular root is similarly deterministic: given a bitmask, the classification (good/ghost/bad) is computable and verifiable.

The fail-open / fail-closed design is explicit: observability paths (telemetry, audit logs) fail open so they never block operations; action paths (C2 commands, swarm manoeuvres) fail closed so nothing ungated escapes. This per-gate explicit choice is documented and tested — not defaulted.

7. ATM Relevance and U-Space Integration

The proposed CNS resilience stack maps directly onto the U-space service framework and the broader ATM architecture:

Table 10: U-space service mapping for Mycelia + Sedenion

U-Space Service	Mycelia Component	Sedenion Contribution
e-registration	Ontology-based identity (IRI per UAV)	—
e-identification	Provenance trail per flight	Triangular root classification of signal origin
Geo-awareness	Milvus GEOMETRY + geo-fast-core	—
Traffic information	SSE streaming + CopMap visualisation	Integrity-checked telemetry (good/ghost/bad flag)
Flight authorisation	Crepe ROE constraints + egress gates	—

Conformance monitoring	DeviationDetector + EVT sentinel	Triangular root anomaly detection; ZDA structural flag
C2 link resilience	SGT mesh self-organisation	16D anti-jamming encoding + threat-type classification

The FLY AI 2026 event's category structure includes **CNS, AIM, MET** and **Trustworthy AI (safety, regulation, certification, cyber)**. This proposal bridges both categories: the sedenion layer is a CNS technology for signal resilience, while the Mycelia platform addresses trustworthy AI through its structural certification approach.

The integration path into existing ATM systems is via the **cartridge boundary**. Just as Mycelia currently isolates SISFRON (military border surveillance) and S1000D (aerospace maintenance) behaviour in separate cartridges, an ATM cartridge would encapsulate U-space-specific ontologies (airspace classifications, separation minima, procedural rules), workflows (flight authorisation, conformance monitoring, contingency procedures), and egress policies. The core engine never learns ATM domain behaviour; it navigates the graph.

8. Roadmap and FLY AI Demonstration

The FLY AI 2026 demonstration will showcase a **fully offline, laptop-deployable** system running the integrated Mycelia + Sedenion stack. The demonstration scenario simulates a U-space surveillance mission with 5 UAVs operating in a GNSS-contested environment:

Table 11: FLY AI 2026 demonstration scenario

Phase	Action	What the Audience Sees
1. Setup	Load SISFRON cartridge; initialise 5-UAV swarm	Dashboard with live COP, UAV positions, mesh topology
2. Normal ops	Swarm patrols sector; C2 via sedenion-encoded mesh	Signal constellation visualisation; triangular root classifies all signals as "good"
3. Jamming event	Simulated GNSS jammer activated	Signals flip to "bad" (no projective structure); mesh re-routes via SGT; no C2 loss

4. Spoofing attempt	Simulated spoofer injects false position	Signals classified as "ghost" (projectively valid but ZD-degenerate); EVT sentinel flags anomaly; Crepe gates action
5. Recovery	Jammer deactivated; swarm resumes normal ops	Full audit trail replay; provenance graph visualisation; triangular root returns to "good"

The demonstration runs entirely on the laptop brought to the room — no cloud calls, no venue network dependency. This air-gap capability is not a demo feature; it is the production deployment mode for the SISFRON cartridge, validated in DIL conditions.

The triangular root is displayed in real time on the dashboard: each inter-UAV signal shows its classification (good/ghost/bad) with the computed mask and its projective dimension. When jamming occurs, the audience sees the classification flip from "good" to "bad" — the signal's basis support loses projective closure. When spoofing occurs, the classification flips to "ghost" — the support remains projectively closed but fails the zero-divisor filter. This provides an **intuitive visual demonstration** of a mathematically rigorous integrity check.

Beyond FLY AI, the research roadmap extends the sedenion layer in three directions:

Table 12: Post-FLY AI research roadmap

Phase	Objective	Deliverable	Timeline
Phase 1 (Q4 2026)	Characterise BER performance under jamming	Simulation benchmark vs. conventional DSSS	3 months
Phase 2 (Q1 2027)	Integrate triangular root into Rust sedenion crate	Production-grade filter with benchmark suite	3 months
Phase 3 (Q2 2027)	Integrate with physical-layer radio (SDR)	Over-the-air test on software-defined radio	6 months
Phase 4 (Q3 2027)	Formal safety case for certification	Assurance argument structured to ED-153	6 months

9. Conclusion

GNSS interference is no longer a theoretical risk — it is an operational reality that EUROCONTROL and EASA have identified as a "significant and evolving challenge" requiring immediate, coordinated action. U-space operations, which depend on GNSS for navigation and C2, are particularly exposed. Current mitigation strategies focus on multi-layer PNT and mesh networking redundancy, but they leave the **signal encoding layer** unhardened.

This proposal presents a dual-layer solution: the **Mycelia platform** provides a certifiable neuro-symbolic runtime for decentralised swarm coordination, with deterministic topology computation, doctrinal gating, and end-to-end provenance; the **Sedenion layer** provides a novel 16D algebraic signal encoding that is structurally resistant to jamming, with non-commutative directional encoding, multi-dimensional spreading, zero-divisor anomaly detection, and the **triangular root structural classifier** that distinguishes three types of signal corruption with exact mathematical guarantees.

The combination addresses the CNS resilience gap at two levels: the sedenion layer hardens the signal against interference while classifying the *type* of threat (brute-force jamming vs. intelligent spoofing), and the Mycelia runtime ensures that every swarm decision is traceable, gated, and auditable. Both layers shift the certification target from stochastic AI models to deterministic algebraic structures — structures that aviation safety thinking already knows how to handle.

The system is production-ready for the control plane (Mycelia has been operational since 2025) and research-validated for the signal plane (the sedenion crate has reproducible test results and benchmark data; the triangular root has a published mathematical proof with verifiable counts). The FLY AI 2026 demonstration will show both layers operating together in a realistic U-space scenario — fully offline, fully traceable, and fully certifiable.

> **Closing Argument**
>
> EUROCONTROL's FLY AI report asks how AI can deliver tangible value in ATM. The answer proposed here is that AI's value is not in replacing human judgment but in creating **verifiable structures** that constrain agency within bounds that humans can audit. The sedenion algebra is one such structure — a 16-dimensional signal manifold with a mathematically proven classification of its 32,767 possible masks into 30 sound, 36 anomalous, and 32,701 invalid. Mycelia is the runtime that certifies navigation across that manifold. Together, they make U-space CNS resilient by design.

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